

**Online Appendix to**

# **Almost a Policy**

**Transmission through Administrative Handoffs**

Chae-Yeon Xon

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This appendix contains proofs, Oregon data construction, additional estimates, and the Monte Carlo exercise. The notation and target definitions follow the main text.

## Appendix A. Proofs and Additional Evidence

### A.1. Proof of the arrival convolution

PROOF OF PROPOSITION 1. Conditioning on  $D_i = \ell$  and aligning the response by the nominal event date gives  $m_h = \sum_{\ell=0}^h f_\ell g_{h-\ell}$ . For any sequence  $f$  with  $f_0 > 0$ , the equation at  $h = 0$  gives  $g_0 = m_0/f_0$ . At horizon  $h \geq 1$ , separating the term with  $\ell = 0$  and dividing by  $f_0$  gives the displayed recursion. The point mass  $f_0 = 1$  yields  $g = m$ . The distinct probability law  $f_0 = f_1 = 1/2$  and the same recursion yield another response sequence with the same  $m$ , establishing multiplicity over the unrestricted real response-sequence class.  $\square$

### A.2. Proof of the loss-acquisition ledger

PROOF OF THEOREM 1. The Markov handoff gives  $I_{j-1} \geq I_j^-$ , hence  $N_{j-1} \subseteq N_j^-$  and  $M_{j-1} \subseteq M_j^-$ . Positive semidefiniteness gives  $\ker(I_j^- + A_j) = \ker I_j^- \cap \ker A_j$ , hence  $N_j \subseteq N_j^-$  and  $M_j \subseteq M_j^-$ . Thus  $\ell_j$  and  $\rho_j$  are nonnegative and

$$\ell_j - \rho_j = \dim M_j - \dim M_{j-1}.$$

Summation telescopes. Under a local reparameterization  $\phi = R\theta$  with invertible  $R$ , null spaces become  $RN$  and the target derivative becomes  $CR^{-1}$ , giving  $(CR^{-1})(RN) = C(N)$ . An invertible recoding preserves the experiment and its Fisher null space. An invertible target transformation maps every  $M$  bijectively to its transformed image. Each dimension difference is therefore invariant.  $\square$

### A.3. Proof of reception regret

PROOF OF COROLLARY 1. For a local decision loss with Hessian  $H$ , the Gaussian local asymptotic risk contribution of the target error is  $\mathcal{R}_{H/2}(I, C)$ . Substitution of  $\Omega = H/2$  into the definitions of  $\Delta_{j,\Omega}^{\text{loss}}$  and  $\Delta_{j,\Omega}^{\text{acq}}$  gives the two changes in the attainable regret bound.  $\square$

### A.4. Proof of minimum upstream repair

PROOF OF THEOREM 2. Set  $K = \ker I$  and  $K_0 = \ker(I + L) = \ker I \cap \ker L$ . On the quotient  $\mathcal{K} = K/K_0$ , define  $\langle [h], [v] \rangle_L = h'Lv$ . This expression is well defined because  $Lk_0 = 0$  for  $k_0 \in K_0$ , and it is positive definite on  $\mathcal{K}$ . The target induces

$$\bar{C} : \mathcal{K} \longrightarrow C(K)/C(K_0), \quad \bar{C}[h] = Ch + C(K_0),$$

with rank  $\dim C(K) - \dim C(K_0) = \ell_j$ .

A  $k$ -coordinate channel  $T_R = R'U$  induces  $\bar{B}_R[h] = R'L^{1/2}h$  on  $\mathcal{K}$ . Its augmented null space is

$$\ker(I + Q_R) = K \cap \ker(R'L^{1/2}).$$

Equality of upstream and repaired target ambiguity implies  $\ker \bar{B}_R \subseteq \ker \bar{C}$ . Rank-nullity then gives  $k \geq \text{rank } \bar{B}_R \geq \text{rank } \bar{C} = \ell_j$ .

For attainment, choose a linear map  $B : \mathcal{K} \rightarrow \mathbb{R}^{\ell_j}$  with  $\ker B = \ker \bar{C}$ . By the Riesz representation theorem on  $(\mathcal{K}, \langle \cdot, \cdot \rangle_L)$ , each coordinate has a vector  $[v_s]$  satisfying  $B_s[h] = v_s' L h = (L^{1/2} v_s)' L^{1/2} h$ . Let the columns of  $R$  be  $L^{1/2} v_s$ . Then  $\ker \bar{B}_R = \ker \bar{C}$ . Since  $K_0$  lies in the channel null space, this equality yields  $C\{\ker(I + Q_R)\} = C(K_0)$ . The constructed channel has  $\ell_j$  coordinates.  $\square$

### A.5. Proof of placement

**PROOF OF COROLLARY 2.** Let  $T$  be any statistic formed from the retained experiment. The data-processing inequality gives  $I_T \leq I$ , hence  $\ker I \subseteq \ker I_T$ . Applying  $C$  gives  $C(\ker I) \subseteq C(\ker I_T)$ , so a retained-data statistic cannot remove any ambiguity created before the receiving stage. Preserving the quotient directions in  $C(\ker I)/C\{\ker(I + L)\}$  therefore requires a channel at or before the handoff. Information generated later is represented by the receiving-stage acquisition term.  $\square$

### A.6. Gaussian information menu

Information contraction gives  $Q \leq L$  for every Gaussian garbling of the lost increment, and Fisher information is positive semidefinite. Conversely, fix  $0 \leq Q \leq L$  and set  $P = L^{\dagger/2} Q L^{\dagger/2}$  on the range of  $L$ . Then  $0 \leq P \leq \mathbf{I}_p$  on that range and  $Q = L^{1/2} P L^{1/2}$ . For an independent  $\varepsilon \sim \mathcal{N}(0, \mathbf{I}_p)$ , observe

$$T = P^{1/2} U + (\mathbf{I}_p - P)^{1/2} \varepsilon.$$

The covariance of  $T$  is  $\mathbf{I}_p$ , its local mean derivative is  $P^{1/2} L^{1/2}$ , and its Fisher information is  $L^{1/2} P L^{1/2} = Q$ . Diagonalizing  $P$  and discarding zero-score coordinates gives score dimension  $\text{rank } Q$ .

### A.7. Proof of minimum precision repair

Let  $B = I + Q$  and  $F = I + L$ , with  $F > 0$ . Normalize the full experiment by setting  $\tilde{B} = F^{-1/2} B F^{-1/2} \leq \mathbf{I}_p$  and  $\tilde{C} = C F^{-1/2}$ . The target is identified under  $B$  when  $\text{range}(\tilde{C}') \subseteq \text{range}(\tilde{B})$ . On this range, every eigenvalue of  $\tilde{B}^\dagger$  is at least one. Hence  $C B^\dagger C' = C F^{-1} C'$  holds exactly when every column of  $\tilde{C}'$  belongs to the unit-eigenvalue space of  $\tilde{B}$ . Transforming

back gives

$$(A.1) \quad (L - Q)F^{-1}C' = 0.$$

Represent any feasible  $Q$  as  $L^{1/2}PL^{1/2}$  with  $0 \leq P \leq \mathbf{I}_p$ . Positive semidefiniteness and (A.1) imply

$$(\mathbf{I}_p - P)L^{1/2}F^{-1}C' = 0.$$

Thus  $P$  acts as the identity on the column space of  $G = L^{1/2}F^{-1}C'$ , which gives  $\text{rank } Q \geq \text{rank } G$ . Setting  $P = P_G$  attains this lower bound and satisfies (A.1), proving the first claim.

For the rank comparison, take  $w$  in the null space of  $G$ . For every  $h \in \ker I$ ,

$$w'Ch = h'FF^{-1}C'w = h'IF^{-1}C'w + h'LF^{-1}C'w = 0.$$

The first term vanishes because  $Ih = 0$ , and the second vanishes because  $Gw = 0$ . Hence  $\ker G \subseteq \{w : w'C(\ker I) = 0\}$ , and rank-nullity gives  $\text{rank } G \geq \dim C(\ker I) = \ell$ ; because  $F > 0$  implies  $\ker F = \{0\}$ .

### A.8. Proof of pilot implementation

Conditional on the pilot, the production channel is a Markov kernel with loading  $R(\tilde{\theta})$ . Quadratic-mean continuity of the score and continuity of the loading give  $Q_{R(\tilde{\theta})} \rightarrow_p Q_{R(\theta_0)}$ . Eigenvalues and singular subspaces separated from zero preserve the selected score rank, while the associated target ambiguity subspace and covariance converge to their limiting values. Integrating over the pilot yields the stated unconditional result.

### A.9. Proof of the conditional-rate release

Write  $\bar{A}_n = K_n/n$  and  $\bar{R}_n = R_n/n$ . The map  $\psi(x, y) = y/x$  has gradient  $(-b/a, 1/a)$  at  $(a, ab)$ . The joint influence function of  $\psi(\bar{A}_n, \bar{R}_n)$  is

$$\frac{A_i(Y_i - b)}{a},$$

which has mean zero, variance  $b(1 - b)/a$ , derivative zero in the  $a$  direction, and derivative one in the  $b$  direction. The multivariate central limit theorem, the delta method, and local asymptotic normality give the displayed limit for  $\hat{b}_n$  under  $(a_n, b_n)$ . Rounding satisfies  $\sqrt{n}|\hat{b}_n - \bar{b}_n| \leq \sqrt{n}\delta_n/2 \rightarrow 0$ , so Slutsky's theorem gives the same limit for the released statistic. The public grid spacing exceeds the  $1/n$  count scale because  $n\delta_n \rightarrow \infty$ . The resulting Gaussian shift has information  $a_0/[b_0(1 - b_0)]$  for  $b$  and zero information for  $a$ . On every compact interior parameter set,  $\mathbb{P}(K_n = 0) = (1 - a)^n$  converges to zero uniformly, so the convention on that event vanishes from the limit.

### A.10. Proof of exact separation and record cost

At every application time  $\tau_r$ , conditional independence gives  $Y_{\tau_r} \sim \text{Bernoulli}(b)$ . The strong Markov property of the iid sequence makes these outcomes independent across  $r$ , so their sum is binomial with parameters  $(m, b)$ . Its score for  $b$  is  $(T_m - mb)/\{b(1 - b)\}$  and has variance  $m/[b(1 - b)]$ . Its law is constant in  $a$ , which gives zero  $a$  score. The waiting increments  $\tau_r - \tau_{r-1}$  are geometric with mean  $1/a$ , hence  $\mathbb{E}[\tau_m] = m/a$ .

For the last claim, write the released law as  $Q_b$ . Its  $b$  score and Fisher information are functions of  $(Q_b, b)$  and therefore have the same value at every  $a$  in the interval. The linked  $n$ -invitation experiment has  $b$  information  $na/[b(1 - b)]$ , which is strictly increasing in  $a$ . Equality between these two information quantities therefore holds at most one value of  $a$ .

### A.11. Proof of the vector frontier

For  $A = I + Q \geq 0$ , the generalized Schur complement gives

$$\begin{pmatrix} A & C' \\ C & V \end{pmatrix} \geq 0 \iff \text{range}(C') \subseteq \text{range}(A), \quad V - CA^\dagger C' \geq 0.$$

Since  $\Omega > 0$ , minimization over  $V$  selects  $V = CA^\dagger C'$  whenever the range condition holds. This proves equivalence with the weighted target-risk program and assigns value  $+\infty$  when the target lacks identification.

The feasible correspondence expands with  $b$ , so the value decreases. If  $(Q_1, V_1)$  and  $(Q_2, V_2)$  are feasible at  $b_1$  and  $b_2$ , their convex combination is feasible at the corresponding convex combination of budgets. The objective is linear in  $V$ , which proves convexity. At  $b = \text{tr}(WL)$ ,  $Q = L$  is feasible and dominates every feasible information matrix, giving the full-retention endpoint.

Under  $I + Q_b > 0$ , differentiation of  $\text{tr}\{\Omega C(I + Q)^{-1} C'\}$  with respect to  $Q$  gives  $-H_b$ . Attach multipliers  $A_b \geq 0$  to  $-Q \leq 0$ ,  $S_b \geq 0$  to  $Q - L \leq 0$ , and  $\lambda_b \geq 0$  to  $\text{tr}(WQ) - b \leq 0$ . Stationarity and complementary slackness give the system in Theorem 4. The envelope formula gives the two directional prices.

### A.12. Proof of stratified link retention

Let  $S_i$  be the sampling indicator for a person in group  $g$ , with  $\mathbb{P}(S_i = 1 \mid g) = q_g$ , independent of the record and the local parameter. If  $s_{ig}$  is the lost score contribution, the observed contribution is  $S_i s_{ig}$ . Its mean is zero and its information is  $\mathbb{E}[S_i s_{ig} s'_{ig}] = q_g L_g$ . Averaging over groups gives  $\sum_g p_g q_g L_g$ . The expected number of retained links divided by the person count is  $\sum_g p_g q_g$ .

### A.13. Scalar specialization

For  $q = 1$ , write the target gradient as  $g$ . Suppose  $I > 0$ , and omit the cap  $Q \leq L$  temporarily. Define  $s = I^{-1/2}g$  and  $H = I^{1/2}WI^{1/2}$ . The cap-free solution at  $b > 0$  is rank one. With  $z_b = (H + b\mathbf{I}_p)^{-1}s$  and  $x_b = \{b/(z_b'Hz_b)\}^{1/2}z_b$ , it is

$$Q_b = I^{1/2}x_bx_b'I^{1/2}, \quad \Phi(b) = g'I^{-1}g - bs'(H + b\mathbf{I}_p)^{-1}s.$$

When  $Q_b \leq L$ , this pair also solves the capped problem. To see the formula, transform  $Q = I^{1/2}RI^{1/2}$  and use

$$s'(\mathbf{I}_p + R)^{-1}s = \max_y \{2s'y - y'y - y'Ry\}.$$

For fixed  $y$ , maximizing  $y'Ry$  over  $R \geq 0$  with  $\text{tr}(HR) \leq b$  produces  $bH^{-1}yy'H^{-1}/(y'H^{-1}y)$ . Substitution and maximization over  $y$  give the displayed expressions.

### A.14. Proofs for rank and frontier inference

For Proposition 2, Weyl's inequality places estimated eigenvalues within  $O_p(a_n)$  of their population values. The threshold conditions recover both null-space dimensions, and Davis–Kahan perturbation bounds give  $\|\widehat{P} - P\| = O_p(a_n/\eta_n)$ . Boundedness of  $\|C\|$  then gives  $\|\widehat{CP} - CP\| = O_p(a_n/\eta_n)$ . Singular-value perturbation,  $a_n/\eta_n = o(c_n)$ , and  $c_n = o(\eta_n)$  recover each image rank, and their difference yields  $\ell_j$ .

For Proposition 3, coverage of  $\mathfrak{C}_{n,j}(1 - \alpha)$  implies that the true primitive belongs to the region with asymptotic probability at least  $1 - \alpha$ , uniformly over the stated class. On that event, the region intersects  $\mathfrak{P}_j(\ell_j)$  at the true primitive, so  $\ell_j \in \mathfrak{R}_{n,j}(1 - \alpha)$ . At separated primitives, consistency and spectral separation exclude every rank class except the true class with probability approaching one.

For Proposition 8, generalized Schur complements place the value problem in a finite-dimensional convex program with compact  $Q$  set. The uniform identification margin bounds feasible covariance matrices over the budget set. Standard value-function sensitivity gives Hadamard directional differentiability. The directional delta method maps  $Z$  through  $\phi'_{b,\beta}$ . The numerical derivative with  $\varepsilon_n \downarrow 0$  and  $\sqrt{n}\varepsilon_n \rightarrow \infty$  consistently estimates that map on bootstrap draws (Fang and Santos 2019; Hong and Li 2018). Applying the same construction jointly over a finite grid yields the maximum-process band.

## Appendix B. Oregon Data and Additional Results

### B.1. Source and construction

The source is the Oregon Health Insurance Experiment Public Use Data archive distributed by the National Bureau of Economic Research. The analysis links the descriptive, initial mail-survey, and state-program files. The arrival sample contains 13,069 selected people with reported receipt, survey-return date, and lottery-notification date; application and approval fields are complete for 29,799 of 29,834 selected people.

Arrival lag is the number of days from lottery notification to survey return, grouped as pre-notification, 0–28, 29–42, 43–55, 56–84, and 85 days onward. The subsequent-outcome panel begins at day 56. Reason shares use 2,569 selected respondents with an outstanding submission and a multi-response reason item. Percentile intervals resample households within lottery draw. The survey records current status at response; the administrative file supplies later outcomes.

For person  $i$  in group  $g$ , the linked log likelihood contribution is

$$A_i \log a_g + (1 - A_i) \log(1 - a_g) + A_i \{Y_i \log b_g + (1 - Y_i) \log(1 - b_g)\}.$$

The two score components are  $(A_i - a_g)/\{a_g(1 - a_g)\}$  and  $A_i(Y_i - b_g)/\{b_g(1 - b_g)\}$ . Their cross moment is zero by iterated expectations. Summation yields  $I_{a,g}$  and  $I_{b,g}$  in Section 5. The conditional-rate release maps the group records to  $\hat{b}_g = \sum_i A_i Y_i / \sum_i A_i$  and rounds this rate to the public grid  $\delta_g = n_g^{-3/4}$ . Proposition 5 gives its local  $b_g$  information and zero root- $n$   $a_g$  information. The fixed-decision release draws an external ordering and stops at application  $m_g = \lfloor n_g/4 \rfloor$ . Its approval count has information  $m_g/[b_g(1 - b_g)]$ , zero  $a_g$  information, and expected scan length  $m_g/a_g$ . On the resulting application-score null space, the derivative of  $d_g = a_g b_g$  is  $b_g$ , which gives the diagonal quotient map.

For a retained fraction  $q_g$ , application information is  $q_g I_{a,g}$ . The delta-method variance of  $\hat{d}_g$  is

$$\frac{b_g^2 a_g (1 - a_g)}{n_g q_g} + \frac{a_g b_g (1 - b_g)}{n_g}.$$

Multiplication by  $N$  and summation over groups gives the empirical frontier. The first-order condition on every uncapped group gives  $q_g \propto b_g \{a_g(1 - a_g)/p_g\}^{1/2}$ ; capped groups are set to one and the remaining budget is reallocated among the remaining groups.

TABLE A1. Reception-stage contrasts in the lottery records

| Lottery-list contrast                          | Application         | Approval   applied  |
|------------------------------------------------|---------------------|---------------------|
| Phone absent minus recorded, English requested | -11.2 [-13.0, -9.3] | 2.4 [0.0, 4.9]      |
| Other language minus English, phone            | 0.3 [-2.0, 2.5]     | -11.6 [-14.4, -8.7] |

*Notes.* Entries are percentage-point differences with household-bootstrap 95% percentile intervals. The records describe contact fields supplied at lottery signup. The contrasts measure stage-specific associations within selected households.

TABLE A2. Approval rates under weighting and draw deletion

| Group                                  | Selected cohort | Lottery-list weighted | Leave-one-draw range |
|----------------------------------------|-----------------|-----------------------|----------------------|
| English requested, phone               | 30.4            | 31.3                  | [29.3, 31.6]         |
| English requested, phone absent        | 26.1            | 26.4                  | [25.2, 27.3]         |
| Other language requested, phone        | 23.3            | 23.5                  | [22.1, 24.0]         |
| Other language requested, phone absent | 25.0            | 23.5                  | [22.0, 27.1]         |

*Notes.* Entries are percentages. Lottery-list weights invert the empirical selection rate within the number-of-household-members stratum. The final column gives the range from repeating the selected-cohort calculation after removing each of the eight lottery draws.

## B.2. Monte Carlo

TABLE A3. Monte Carlo coverage near a frontier kink and a weak rank

| Procedure and design              | Nominal | Coverage | Mean interval/set size |
|-----------------------------------|---------|----------|------------------------|
| Numerical delta, smooth budget    | 0.95    | 0.932    | 3.101                  |
| Ordinary bootstrap, smooth budget | 0.95    | 0.938    | 3.112                  |
| Numerical delta, first cap        | 0.95    | 0.941    | 3.298                  |
| Ordinary bootstrap, first cap     | 0.95    | 0.942    | 3.289                  |
| Rank set, separated               | 0.95    | 1.000    | 1.000                  |
| Rank set, local fourth direction  | 0.95    | 1.000    | 1.837                  |
| Rank set, rank-three boundary     | 0.95    | 0.984    | 1.984                  |

*Notes.* The design uses 100,000 observations, four group shares and transition probabilities calibrated to the Oregon sample, 1000 Monte Carlo replications, and 499 Gaussian score draws per replication. The budget is the expected share of person links retained. The first-cap budget is the population value at which one group reaches full retention. Frontier size is average interval length; rank size is average confidence-set cardinality.

## References

- Fang, Zheng and Andres Santos**, “Inference on Directionally Differentiable Functions,” *The Review of Economic Studies*, 2019, 86 (1), 377–412.
- Hong, Han and Jessie Li**, “The Numerical Delta Method,” *Journal of Econometrics*, 2018, 206 (2), 379–394.